MILITARY SPECIFICATION

CARTRIDGE, 105MM, HE M1, (MOD) LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This document covers the loading, assembling and packing for one type of HE Cartridge designated as M1 for use with the 105MM Howitzer M137.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-Ammunition and Special Weapons, General Specification for
MIL-C-45662	-Calibration System Requirements
MIL-P-46565	-Primer Percussion, M28A2 and M28B2, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-414	-Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-453	-Inspection, Radiographic
MIL-STD-1168	-Lot Numbering of Ammunition
MIL-STD-1235	-Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

9276004	-Packing and Marking for Container, Ammo Fiber PA-71, For 105MM How. Ammunition With Fuze (USAF Use Only)

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9276005 -Container, Ammo, Fiber PA-71 For 105MM How. Ammo W/Fuze (USAF Use Only)
9277353 -Packing and Marking for Box, Pallet Type, Wirebound for 105MM How. Ammo W/Fuze (USAF Use Only)
9278402 -Box, Pallet Type, Wirebound, Packing, Ammo For 105MM W/Fuze (USAF Use Only)
9278792 -Cartridge 105MM: HE, M1 (MOD)

PUBLICATIONS

ARMY

ET-9278792-Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Material and parts shall be in accordance with applicable drawings and specifications. (See 6.9)

3.1.1 Clean explosive scrap.-Clean explosive scrap produced in regular operations may be remelted and reused provided that it complies with the requirements for the grade of high explosive specified, except for form. Scrap from floor sweeping, ventilation and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used. (See 6.7)

3.2 Cartridges.-The cartridges shall comply with all requirements specified on Drawing (dwg.) 9278792 and with all requirements specified in applicable specifications.

3.3 Cavitiation in high explosive charge.

3.3.1 Projected cavities.-Cavities having a projected length of 1/32 inch (in.) or less shall be disregarded. Cavities within the explosive charge shall not exceed the requirements specified in Table I, when determined as specified in 4.3.1.

TABLE I

	(Segments (see Figure 1)			
	A	B	C	D
Sum of projected areas of the cavities, excluding pipes, cracks and annular rings, square inch (sq. in).	*1/64	1/8	1/2	1/2

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Projected length of any cavity, excluding pipes, cracks, and annular rings (in.).	1/8	1/2	1/2	3/4
Piping cavities, maximum (max.) projected area (sq. in.).	0	1/2	1/2	-
Piping cavities, max. projected width (in.)	0	1/2	1/2	-
Cracks, max. projected width (in.).	1/32	1/32	1/32	-
Annular rings, max. projected width (in.).	0	0	1/4	0

*If the length of the largest cavity is 1/16 inch or less, the maximum total projected area may be 1/20 square inch.

3.3.2 Surface cavitation.-Cavities formed due to chipped or broken explosive from the sidewalls of the fuze well shall not extend in aggregate around more than 1/4 of the circumference. Cavities in the base of the fuze well shall not be cause for rejection provided the sum of the areas of the cavities is not greater than 80 percent of the area of the fuze well base and no individual point extends more than 1/4 inch below the maximum depth of the fuze well permitted by the drawing, and provided no point extends above the flat surface on which the liner would rest. (Cavities having a maximum dimension of 1/32 inch or less shall be disregarded in this computation as shall voids within the applicable forward region of the fuze well prescribed by dwg. 9211610).

3.3.3 Porous areas.-Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.3.4 Cracked charges.-Not more than two transverse cracks shall be permitted in any charge, and not more than one transverse crack shall be permitted in Segment A.

3.3.5 Base separation.-There shall be no separation between the shell base and explosive charge.

3.4 Proving ground.-The cartridge shall comply with the following requirements when tested as specified in 4.3.2.

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3.4.1 Muzzle velocity.-The corrected muzzle velocity of the projectile at ambient temperature shall be not more than 1645 feet per second nor less than 1595 feet per second and the standard deviation of the lot shall be not greater than 8 feet per second (2.5 meters per second).

3.4.2 Premature bursts and security of parts.-There shall be no premature burst or indication of metal parts failure in the gun bore or in flight before impact.

3.4.3 Safety.-There shall be no premature burst or indication of metal parts separation in the gun bore or in flight when fired with propellant adjusted to give a pressure of 42,000 pounds per square inch (psi) plus or minus 1500 psi.

3.5 Workmanship

3.5.1 Cartridge.-All manufacturing processing and assembly operations of the cartridge shall be correctly performed in a thoroughly workmanlike manner. Parts shall be free of burrs, chips, dirt, cracks, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

3.5.1.1 Loading of high explosive.-The interior surface of the projectile shall be dry when the high explosive is poured into the projectile. Solid explosives shall not be present in the projectile before loading or be added to the projectile during pouring or cooling.

3.5.1.2 Painting and marking.-The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked when such painting and marking has been marred or obliterated.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent' authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revision, same specifications and same specification revisions. Changes to the process, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection

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inspection equipment,

lots of components or subassemblies, other than the items of delivery shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Cartridge cases of one lot interfix number from one manufacturer.

b. Projectile metal parts of one lot interfix number from one manufacturer.

c. Propelling charges from not more than one lot.

d. Loaded primers of one lot interfix number from one manufacturer.

e. Supplementary charges from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification and same specification revision.

f. Fuze well liners from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification and same specification revision.

g. TNT of the same type from one lot interfix number from one manufacturer; or Composition B of the same grade from one lot interfix number from one manufacturer which contains TNT from not more than one lot interfix number from one manufacturer.

h. Loaded projectile of same weight zone.

4.2.2 Examination.-Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Projectile loading assembly, prior to loading explosive (see dwg. 9211610 or 9278812 covering a detail of dwg. 9278792).

Categories	Defects	Method Of Inspection
Critical:		
	1. Foreign matter in charge cavity.....	Visual
	2. Unsolidified pool of paint in bottom of cavity.....	Visual

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Major:	AQL 0.40 percent	
101.	Interior protective coating with bare spot or blister	Visual
Minor:	AQL 0.65 percent	
201.	Evidence of poor workmanship (see 3.5)	Visual

4.2.2.2 Projectile loading assembly, prior to assembly of liner (see dwg. 9211610 or 9278812 covering a detail of dwg. 9278792).

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Depth of fuze well, maximum, (max.)	Gage CD
102.	Foreign matter in fuze well	Visual
Minor:	AQL 1.5 percent	
201.	Diameter of fuze well, max	Gage CD
202.	Surface cavitation excessive(applicable to deep cavity) (see 3.3.2)	Visual
203.	Evidence of poor workmanship (see 3.5)	Visual

4.2.2.3 Projectile loading assembly prior to assembling supplementary charge (see dwg. 9211610 or 9278812 covering a detail of dwg. 9278792).

Categories	Defects	Method Of Inspection
Critical:		
1.	Liner missing	Visual
Major:	AQL 0.40 percent	
101.	Depth of bottom of liner minimum (min)	Gage CD
102.	Concentricity of inside diameter of liner with pitch diameter of thread including perpendicularity with face of body	Gage CD
Minor:	AQL 1.5 percent	
201.	Liner damaged	Visual
202.	Foreign matter in liner cavity including explosive	Visual
203.	Evidence of poor workmanship (see 3.5)	Visual

4.2.2.4 Projectile loading assembly, prior to assembly of closing plug (see dwg. 9211610 or 9278812 covering a detail of dwg. 9278792).

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Categories	Defects	Method Of Inspection
Critical:		
	1. Pad on supplementary charge missing	Visual
	2. Supplementary charge missing	Visual
	3. Projectile zone weight incorrect by two or more zones	Balance CD
Special 100 percent inspection (see 6.2) - Explosive on exterior of projectile		Visual
Major:	AQL 0.40 percent	
	101. Projectile zone weight incorrect by one zone	Balance CD
	102. Supplementary charge inverted	Visual
Minor:	AQL 2.5 percent	
	201. Supplementary charge damaged (applicable to deep cavity)	Visual
	202. Spacer missing or damaged (when applicable)	Visual
	203. Rotating band damaged	Visual
	204. Protective coating with total damaged area in excess of 1/4 square inch	Visual
	205. Marking missing, misleading or unidentifiable (when applicable)	Visual
	206. Evidence of poor workmanship (see 3.5)	Visual

4.2.2.5 Primer, prior to assembly to case (see dwg. 8838130 covering a detail of dwg. 9278792).

Categories	Defects	Method Of Inspection
Critical:		
	1. One or more flash holes missing	Visual
Major: None defined.		
Minor: None defined.		

4.2.2.6 Cartridge case, prior to assembling propelling charge (see dwg. 7548409 or 7548025 covering a detail of dwg. 9278792).

Categories	Defects	Method Of Inspection
Critical: None defined.		

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Major: AQL 0.40 percent
101. Primer missing or damaged.....Visual

Minor: AQL 1.0 percent
201. Foreign matter in interior of case.....Visual
202. Evidence of poor workmanship (see 3.5).Visual

4.2.2.7 Propelling charge, prior to assembly to cartridge case (see dwg. 9276669 covering a detail of dwg. 9278792.

Categories	Defects	Method Of Inspection
Critical:		
Major:	AQL 0.40 percent	
101.	Any opening in bag large enough to permit escape of propellant.....Visual	
Minor:	AQL 0.65 percent	
201.	Evidence of poor workmanship.....Visual	

4.2.2.8 Cartridge Case with propellant charge prior to assembling to projectile (see dwg. 754809 or 7549025 covering a detail of dwg. 9278792).

Categories	Defects	Method Of Inspection
Critical: None defined		
Major:	AQL 0.40 percent	
101.	Foil side of bag not facing primer.....Visual	
Minor: None defined		

4.2.2.9 Cartridge assembly (see dwg. 9278792).

Categories	Defects	Method Of Inspection
Critical:		
1.	Primer above flush.....Gage	CD
Major:	AQL 0.65 percent	
101.	Torque test failure of plug or fuze....Test	CD
102.	Any component missing.....Visual	
103.	Fuze not fully seated.....Visual	

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Minor: AQL 1.5 percent

201.	Exterior coating of shell with an aggregate of bare spots in excess of 1/4 square inch	Visual
202.	Marking misleading, unidentifiable or missing including incorrect color (except zone weight)	Visual
203.	Plug not fully seated	Visual
204.	Evidence of poor workmanship (see 3.5)	Visual

4.2.2.10 Unsealed fiber container (see dwg. 9276004 and 9276005).

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Packing component missing	Visual
Minor:	AQL 1.5 percent	
201.	Cartridge cannot be removed by hand	Manual
202.	Container cap cannot be removed by hand	Manual
203.	Glue or asphalt on cartridge	Visual
204.	Packing components improperly assembled	Visual

4.2.2.11 Sealed fiber container (see dwg. 9276004 and 9276005).

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	AQL 0.65 percent	
101.	Container cut or damaged through all asphalt impregnated layers	Visual-Manual
102.	Metal end loose or distorted	Visual-Manual
103.	Tape incomplete or badly wrinkled	Visual
Minor:	AQL 2.5 percent	
201.	Tear tab length, min	Gage
202.	Contents loose	Manual
203.	Gap between cover and body container exceeding 1/8 inch	Visual
204.	Cuts, scuffs or gouges in outer layers	Visual
205.	Marking misleading, missing or unidentifiable	Visual

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4.2.2.12 Sealed pallet type wirebound (see dwg. 9278402 and 9277353).

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	AQL 0.65 percent	
101.	Box damaged exposing contents	Visual
102.	Hardware or strapping missing, broken or loose	Visual-Manual
103.	DOD symbol missing or incorrect	Visual
104.	Board broken	Visual
Minor:	AQL 1.5 percent	
201.	Strapping or hardware improperly assembled	Visual-Manual
202.	Handle missing or insecure	Visual-Manual
203.	Car seal missing, unsealed or improperly positioned	Visual
204.	Marking misleading, missing or unidentifiable	Visual

4.2.3 Testing

4.2.3.1 Check test for possible deterioration of primer, Percussion, M28, (see specification MIL-P-46565)-If the total elapsed time between original acceptance of the M28 primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the M28 primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-P-46565 immediately before the M28 primer lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridges (see 6.4).

4.2.3.2 Cavitation (see 3.3 except 3.3.2)-The projectiles shall be radiographed 100 percent and any projectile which fails to meet the requirements shall be classed defective and removed from the lot.

4.2.3.3 Caked explosive in threads (see dwg. 9211610) Minor defect.-The sampling plan for this defect shall be a GSP-2 plan in accordance with MIL-STD-1235 with an AQL of 0.65 percent, inspection level II. The projectile shall be inspected visually for this defect prior to assembly of the liner.

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4.2.3.4 Liner removable by hand (see dwg. 9211610 or 9278812) Minor defect.-The sampling plan for this defect shall be a CSP-2 plan in accordance with MIL-STD-1235 with an AQL of 0.65 percent, inspection level II. The projectile shall be inspected manually for this defect.

4.2.3.5 Fuze and Plug not fully seated (see dwg. 9276632).- The sampling plan for this defect shall be CSP-2 plan in accordance with MIL-STD-1235 with an AQL of 0.65 percent, inspection level II.

4.2.3.6 Profile and Alignment (see dwg. 9278792)-Major defect. The cartridge, including those which had the bourrelet repainted shall be gaged 100 percent for profile and alignment. Any cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot.

4.2.3.7 Bullet pull (see dwg. 9278792).-The sampling plan for this test shall be in accordance with MIL-STD-414, Section B, Part I. Table B1, Code Letter F, and AQL of 1.5 percent. CD

4.2.3.8 Proving ground.(See 6.5).

4.2.3.8.1 Muzzle velocity see 3.4.1 Major defect

4.2.3.8.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan shall be in accordance with Standard MIL-STD-414, Section B, Table B-3, Code Letter I and AQL 1.0 percent. In addition, if the standard deviation of the muzzle velocity multiplied by the factor 0.75 exceeds the applicable lot requirement, the lot shall be rejected.

4.2.3.8.1.2 After the three consecutive lots have met the criteria of 4.2.3.8.1.1, the sampling plan shall be in accordance with Standard MIL-STD-414, Section B, Table B-3, Code letter F and AQL of 1.0 percent. In addition, if the standard deviation of the muzzle velocity multiplied by the factor 0.65 exceeds the applicable lot requirement, the lot shall be rejected.

4.2.3.8.2 Premature bursts and security of parts (see 3.4.4) Critical defect.-If any of the sample projectiles used in any of the tests fail to comply with the applicable requirements, the lot shall be rejected.

4.2.3.8.3 Safety. (see 3.4.3) - Critical defect.

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4.2.3.8.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements, fifteen sample cartridges shall be randomly selected from each lot for this test. If any projectile fails to comply with the requirements, the lot shall be rejected.

4.2.3.8.3.2 After three consecutive lots have met the criteria of 4.2.3.8.3.1, five sample cartridges shall be randomly selected from each lot for this test. If any projectile fails to comply with the requirements, the lot shall be rejected.

4.2.4 The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.2.2 and 4.2.3 or in ET-9278792 (see 6.3).

4.2.4.1 Inspection Equipment Capability and Accuracy.-All inspection equipment and facilities shall be of adequate quality and quantity to perform properly and accurately all the required inspections, measurements and tests. The inspection equipment shall be maintained in satisfactory condition for the continued proper performance of these functions and shall be calibrated in accordance with the provisions of MIL-C-45662.

4.2.4.2 Government Rights to Documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3 Test Methods and Procedures

4.3.1 Determination of Cavities.-The sample projectiles shall be subjected to radiographic examination. The radiographic beam shall be located in a plane which is perpendicular to the longitudinal axis of the projectile and tangent to the base of its internal cavity. The outline of the explosive charge shall be clearly defined at all points. Examination of the radiograph shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks, or other extraneous features should project into the region of explosive under inspection, with the exception of specified penetrameters and radiation screening required to reduce film densities to readable levels in the thinner areas of the shell. Such screening should be free of any radiographically

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observable features in the projected area under inspection. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453 (see 6.8). In addition the sensitivity of the radiographic technique shall be 2.0 percent and the minimum photographic density of the film used shall be no less than 1.25 at every point within the projected area of explosive, including the base and rotating band area, with the maximum density no greater than to permit adequate inspection over the entire area of interest, as determined in accordance with 4.3.1.1 and 4.3.1.2. In the event an original radiograph leaves doubt as to its acceptability, the projectile shall be reradiographed 90 degrees from the original exposure. Acceptability shall be determined on the basis of the film giving the largest projected area of defect.

4.3.1.1 Check of sensitivity of radiographic technique. The sensitivity of radiographic technique shall be determined by means of penetrameters fabricated as specified herein. The penetrameters shall be placed on the radiographic source side of the projectile so that the 2T holes are at the points shown in Figure 2. Three penetrameters for each projectile shall be used. When radiographing a number of similar projectile the frequency of sensitivity check shall be no less than one in each fifty radiographs and on a rotating ring, the penetrameters shall be used at two stations for each complete rotation of the ring. The sensitivity of the radiographic technique shall be considered satisfactory when the image of the pentrameters, including the 2T holes, are sharply defined on the radiograph.

The penetrameters shall be made of steel having an approximate specific gravity of 7.8. The penetrameters shall be located as shown on Figure 2. For further details on construction of the penetrameter see MIL-STD-453.

4.3.1.2 Check of photographic density of film.-The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of the explosive charge. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. To judge densities it is recommended that standards be used (6.6). Comparision of the film with standards may be made visually, however, the use of a densitometer is recommended.

4.3.2 Proving ground.-These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The sample cartridges shall be fired with standard propelling charges in a M137 gun. Observation shall be made for conformance to all requirements. (See 6.5).

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4.3.2.1 Muzzle velocity and functioning.

4.3.2.1.1 Without fuze.-The sample cartridges shall be assembled with live or dummy fuzes and fired for muzzle velocity at zone 7.

4.3.2.1.2 Fuze, M557.-For lots with 25 sample cartridges, 15 of the fuzes shall be set "super quick" and 10 set "delay". For lots with 10 sample cartridges, 5 shall be set "super quick" and 5 "delay". The cartridges shall be fired at zone 7 with impact on firm ground.

4.3.2.2 Safety.-The sample cartridges shall be fired with propelling charges adjusted to give a pressure of 42,000 psi-plus or minus 1500 psi in a weapon with M103 tube. In addition, for the first three production lots from each manufacturer (loading facility), a smear camera shall be used to observe the projectile in flight in order to record any possible malfunctioning. (Cartridges furnished without fuzes shall be assembled with live or dummy fuzes for this test.)

4.3.3 Bullet pull.-The security of the crimp of the cartridge case to the projectile shall be determined by applying a load in the axial direction with a rate of travel of the testing machine head of 0.100 to 0.125 inch per minute. Observation shall be made for compliance with the applicable requirement.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level a.-The cartridges shall be preserved and packaged in accordance with dwg. 9276005.

5.2 Packing

5.2.1 Level A.-The fiber container shall be packed in accordance with dwg. 9278402.

5.3 Marking.-The pallet marked shall be in accordance with 9277353. The fiber container shall be marked in accordance with dwg. 9276004.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the title, number and date of this specifications.

6.2 Special Defects

6.2.1 Definition.-A special defect is a defect that judgement and experience indicate could, depending on the degree of variance from the design requirement, result in

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hazardous or unsafe conditions for individuals using, maintaining or depending upon the product.

6.2.2 Disposition.-Upon investigation and analysis of lots exhibiting special defects by the responsible technical activity a recommendation of acceptance or rejection shall be made to Director of Quality Assurance, U.S. Army Munitions Command. Final acceptance of such lots on waiver shall be made only by the Director of Quality Assurance, U.S. Army Munitions Command.

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types-Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs, if available, are designated by drawing numbers (on the Equipment Lists referenced on the Equipment Tabulation) or (on the Equipment Lists referenced on the Equipment Tabulation) or (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, which established the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is catalogued and available for purchase by the general public.) Contractor designs shall include appropriate operating instruction, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations

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and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number, paragraph number and defect number if such number has been assigned shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Cost of check test.-The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.5 Submission of test data.-In addition to the normal distribution of records, when the cartridge is procured by the Department of the Army, one copy of all ballistic test data will be forwarded to the following:

Commander
Picatinny Arsenal
Dover, New Jersey
ATTN: SMUPA-QA-A-A and SMUPA-AD-E-A

6.6 Photographic densities.-"American Standard for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York has been found satisfactory as a source of information to filter densities.

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6.7 Clean explosive scrap.-Clean explosive scrap is defined as scrap formed in kettles, pails, risers and loading machines used in the pouring and cooling operations, and scrap formed during the drilling operation.

6.8 Radiographic procedures.-Radiographic procedures and sample radiographs shall be submitted to the Commander, Picatinny Arsenal, Dover, New Jersey ATTN: SMUPA-QA-A-A for review and approval upon implementation of this document and subsequently at the end of each 6 months period of production.

6.9 When TNT shows definite frothing tendencies (which may cause production of unsatisfactory charges) 0.1 percent maximum of sorbitan trioleate in accordance with Specification JAN-S-547 may be used. Projectile lots loaded with TNT which has been so treated should have data cards annoted accordingly.

Preparing Activity:
Army-MU

Custodian:
Army-MU

Project Number: 1315-A004

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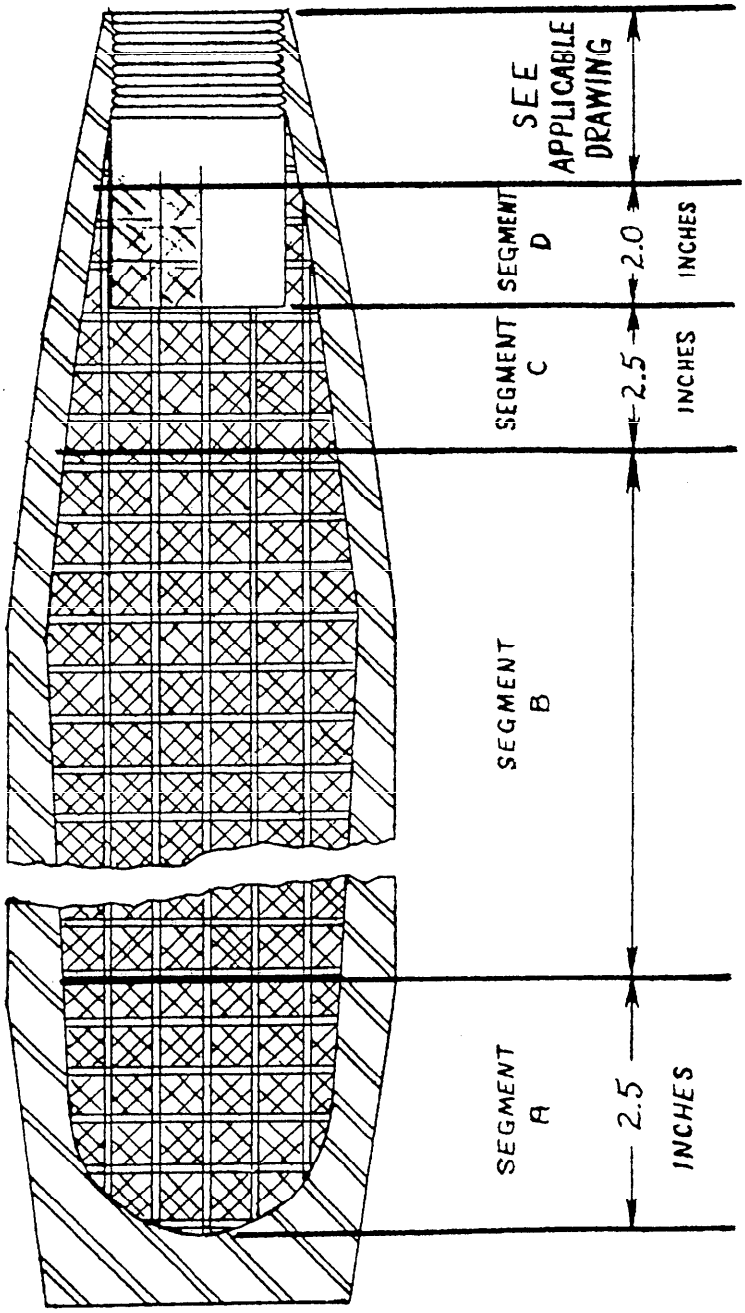
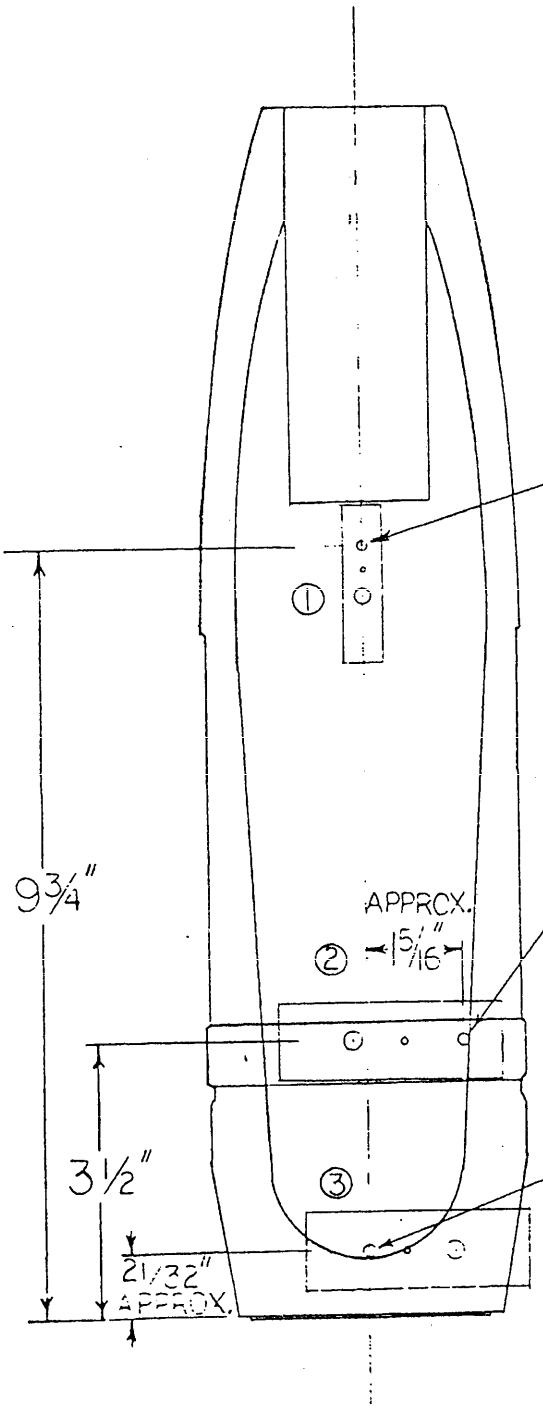


FIGURE 1

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PROJECTILE 105MM, M5, M1
RADIOGRAPHIC PENETRAMETER REQUIREMENTS



Penetrameters Constructed Per MIL-STD--53
Dimensions shown are for location of center of 2T holes.

PENETRAMETER FOR 1.5" STEEL
.75 Total steel wall plus
3.375" explosive.
1.5" equivalent thickness
2% penetrameter thickness =.030"
2T Hole diameter =.060"

PENETRAMETER FOR 2.7" STEEL
2.57" Total steel wall plus
.82" explosive at 1/16" from tangent point.
2.74" equivalent thickness
2% penetrameter thickness =.054"
2T Hole diameter =.108

PENETRAMETER FOR 3.1" STEEL
3.0" Total steel wall plus
.75" explosive at 1/16" from tangent point.
3.15" equivalent thickness
2% penetrameter thickness =.062"
2T Hole diameter = .12-"

FIGURE 2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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FOLD

Department of the Army
Headquarters, Picatinny Arsenal
Dover, New Jersey 07801

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-QA-A
Dover, New Jersey 07801

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